10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 25 — Biochemistry

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

25.1 Which two functions are performed by proteins?

(i) They are needed to build new cells and repair the injured ones, because proteins form skin, hair, muscle, blood and almost all the organs of animals.

(ii) They work as enzymes which catalyse the biochemical reactions of the body; hormones such as insulin and haemoglobin, which carries oxygen in the blood, are also proteins.

25.1 Is milk a fat, a protein or both?

Milk contains both. It contains animal fat (butter and ghee are made from milk) as well as the protein casein, along with water, lactose and minerals. That is why milk is called a complete food.

25.2 What is a balanced diet?

A balanced diet is the diet which contains all the nutrients — carbohydrates, lipids, proteins, vitamins, minerals, fibre and water — in proper amounts. It provides fuel for healthy growth and development, keeps the weight normal, supports brain function and a strong immune system, and prevents diseases such as heart disease, diabetes, some cancers and osteoporosis.

25.2 Name any three nutrients.

Carbohydrates, lipids (fats) and proteins are three nutrients. Vitamins and minerals are also nutrients which our body needs regularly to maintain health and prevent diseases.

25.3 Name any two sources of starch.

Starch is found in foods like potatoes and rice; grains such as wheat and maize are also rich sources of starch. Cellulose, the other polymeric carbohydrate, is obtained from plant materials such as wood and cotton.

25.3 Are all carbohydrates equally beneficial for our health?

No. All carbohydrates give energy, but they are not equally beneficial. Starch is a polymeric carbohydrate which gives energy slowly and also provides fibre, while cellulose helps in digestion. On the other hand, table sugar (sucrose) and other simple sugars give energy very quickly but contain no fibre, vitamins or minerals, and eating too much of them is harmful. So a diet should contain more starch and fibre and less refined sugar.

25.4 What are hormones made of?

Hormones are protein in nature. For example, insulin is a hormone which controls the level of sugar in the blood stream. Hormones are made of chains of amino acids joined by peptide linkages, so chemically they are proteins or peptides.

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25.4 What is denaturation of protein?

Complex changes occur in proteins when they are subjected to heat, a change in pH or an attack by oxidising and reducing agents. This phenomenon is called denaturation of proteins. During denaturation the three dimensional structure of the protein is disturbed, so it loses its natural shape and its biological activity. A familiar example is the frying or boiling of an egg, which involves the denaturation of egg albumin, the principal component of egg white.

25.5 What is the role of lipids in our body?

Lipids act as major energy reserves of our body and provide, on the average, twice as much energy as carbohydrates or proteins. They also provide insulation and protection for the vital organs of our body. Animal fats are stored in adipose tissue cells, while vegetable oils are present in the seeds and roots of plants.

25.5 Which is an unsaturated fatty acid?

Oleic acid is an unsaturated fatty acid. It is present in olive oil and its chain contains a carbon–carbon double bond, that is why it is not saturated. Lauric acid, stearic acid and palmitic acid contain only single bonds, so they are saturated fatty acids.

25.6 Which vitamin is present in lemon?

Lemon contains vitamin C, which is also called ascorbic acid. It is found in citrus fruits, green pepper and tomatoes, and its deficiency in our body causes scurvy.

25.6 Compare the roles played by vitamins and carbohydrates in our body.

Vitamins	Carbohydrates
They are organic substances which do not provide energy.	They are the main source of energy for the human body.
They are needed in very small amounts to maintain normal health and growth.	They are needed in large amounts; a healthy adult needs about 225–325 g per day.
They function as catalysts for the biological processes going on in our body.	They are broken down to give energy and also help the body to use protein and fat efficiently.
Their deficiency causes diseases such as night blindness, rickets and scurvy.	Their deficiency makes the body weak because the body gets no fuel.

25.7 What is the function of RNA?

The function of RNA is to read, decode and use the information stored in DNA to make proteins. DNA stores the information and passes it on to RNA at the proper time, while RNA molecules are much smaller and are mostly found outside the cell nucleus.

25.7 What are the basic differences between DNA and RNA?

DNA	RNA
Its full name is deoxyribonucleic acid.	Its full name is ribonucleic acid.
Molecules of DNA are very large; a DNA molecule may be up to 12 cm long when stretched out.	RNA molecules are much smaller than those of DNA.
It is mostly found in the nucleus of the cell.	It is mostly found outside the cell nucleus.
Its function is to store information and pass it on to RNA.	Its function is to read, decode and use the information to make proteins.

Exercise (Solved)

A. Multiple Choice Questions

1. Functional groups present in fructose are:

(a) Hydroxyl and aldehyde (b) Hydroxyl and ketone
(c) Hydroxyl and ester (d) Hydroxyl and amide

Answer: (b) — Fructose contains a number of hydroxyl groups along with a ketonic functional group, so it is a ketohexose. Glucose, on the other hand, contains hydroxyl and aldehyde groups, so it is an aldohexose.

2. Which functional group is present in a fat?

(a) Carboxyl group (b) Ester group
(c) Carbonyl group (d) Hydroxyl group

Answer: (b) — Fatty acids react with glycerol to give a triester which is called a fat. The triester of glycerol is also called a triglyceride, and it contains ester linkages.

3. Which function is not performed by proteins in our body?

(a) They serve as chemical messengers. (b) They serve as storage of energy-rich fuel.
(c) They store genetic information. (d) They serve as catalysts for biochemical reactions.

Answer: (c) — Storage of genetic information is the function of nucleic acids (DNA), not proteins. Proteins work as chemical messengers (hormones), as catalysts (enzymes) and as a source of energy, but they do not store genetic information.

4. Which functional groups react to produce a peptide linkage in proteins?

(a) Carboxyl group and hydroxyl group (b) Carboxyl group and amino group
(c) Hydroxyl group and amino group (d) Ester group and amino group

Answer: (b) — An amino acid has a carboxyl group (–COOH) and an amino group ($–NH_2$). When two amino acids join, water is eliminated from these two groups and a peptide linkage is formed.

5. The test performed to detect cancer in early stages is:

(a) Genetic test (b) Urine test

(c) Galleri test (d) Electrolyte test

Answer: (c) — The Galleri test is a specific type of blood test which uses DNA sequencing to find the characteristic patterns which show whether a person is suffering from cancer or not.

6. Fatigue and depression are caused by deficiency of:

(a) Vitamin A (b) Vitamin C

(c) Vitamin D (d) Vitamin B1

Answer: (d) — The deficiency of vitamin B_1 leads to fatigue and depression, while the lack of vitamin B_2 causes inflammation of the lips and dryness and burning of the eyes.

B. Short Answer Questions

25.1 What is the difference between glucose and fructose?

Glucose	Fructose
It is also called grape sugar or dextrose.	It is the sugar of sweet fruit juices and honey.
It contains an aldehyde group, so it is an aldose (aldohexose).	It contains a ketonic group, so it is a ketose (ketohexose).
Its open chain structure is $CHO–(CHOH)_4–CH_2OH$.	Its open chain structure is $CH_2OH–CO–(CHOH)_3–CH_2OH$.
Both have the same molecular formula $C_6H_{12}O_6$, so they are isomers of each other.	Both have the same molecular formula $C_6H_{12}O_6$, but their structures are different.

25.2 Give two examples of denaturation of proteins.

(i) The frying or boiling of an egg denatures the egg albumin, which is the principal component of egg white, and the white part becomes solid.

(ii) When milk becomes sour, the acid formed in it changes the pH and the milk protein casein is denatured, so the milk curdles.

25.3 What are the sources of vitamin D?

Vitamin D may be obtained from butter and fish liver oils. Our skin can also prepare vitamin D directly in the presence of sunlight. This vitamin is related to normal teeth and bone development, and its deficiency causes a disease called rickets.

25.4 What is the role of enzymes in our body?

Enzymes are proteins which act as catalysts. They speed up the biochemical reactions taking place in our body without being used up in the reaction. In this way the digestion of food and all other chemical changes of the body take place at body temperature.

25.5 Which commercial products are obtained from oils and fats?

Soap is prepared by boiling animal fat with an alkali, so soap is a mixture of sodium salts of long chain fatty acids. Oils and fats are also used to make cooking oil, ghee, butter, margarine, candles, cosmetics and paints, and glycerol obtained from fats is used in medicines and cosmetics.

25.6 Which vitamin helps in blood clotting?

Vitamin K helps in the clotting of blood. When a wound occurs, this vitamin helps the blood to clot and stops the bleeding. Green leafy vegetables are a good source of vitamin K.

C. Constructed Response Questions

25.1 What is the function of DNA?

DNA (deoxyribonucleic acid) stores the hereditary information of the cell and passes it on to RNA at the proper time. In this way DNA controls the development, growth and reproduction of all forms of life. DNA molecules are very large and are mostly found in the nucleus of the cells.

25.2 Write down three important functions played by proteins in our body.

(i) Building and repair: proteins are needed to build new cells and repair the injured ones; fibrous proteins are found in muscles, hair, nails, skin and connective tissues.

(ii) As enzymes: enzymes are protein in nature and they catalyse (speed up) the biochemical reactions taking place in our body.

(iii) As hormones and carriers: hormones such as insulin control the sugar level of the blood, and haemoglobin, which carries oxygen in the blood and delivers it to different parts of the body, is also a protein.

25.3 Fatty acids serve as building blocks of lipids just like amino acids are for proteins. Comment on this statement.

The statement is correct. Most lipids are composed of simpler substances called fatty acids, which are long chain carboxylic acids containing from 12 to 28 carbon atoms. Just as thousands of proteins are built by joining only twenty amino acids through peptide linkages, the lipids are built from fatty acids. For example, lauric acid is obtained from coconut oil, stearic acid from animal fats, palmitic acid from palm oil and oleic acid from olive oil. Fatty acids react with an alcohol called glycerol to give a triester which is called a fat (triglyceride); tristearin is formed from stearic acid and glycerol. Thus fatty acids are the units of which lipids are made, just as amino acids are the units of proteins.

25.4 Write down about the sources of vitamin C. Which diseases are caused by its deficiency?

Vitamin C, also called ascorbic acid, is found in citrus fruits such as lemons and oranges, in green pepper and in tomatoes. Its deficiency in our body causes a disease called scurvy, in which the gums bleed and the healing of wounds becomes slow. So we should take fresh fruits and vegetables in our daily diet to get enough vitamin C.

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25.5 Write down the names of two polymeric carbohydrates.

Starch and cellulose are the two important polymeric carbohydrates. Both have the common formula $(C_6H_{10}O_5)_n$, which shows that many simple sugar units are joined together to form these big molecules. Starch is found in foods like potatoes, rice and grains, while cellulose is obtained from plant materials such as wood and cotton.

D. Descriptive Questions

25.1 What are the sources of fructose, maltose and lactose?

Fructose: it is present in sweet fruit juices and honey.

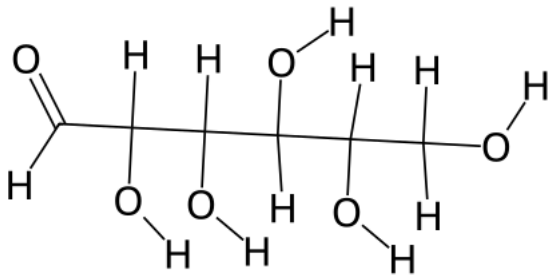
Maltose: it is found in breads and other cereals.

Lactose: it is present in milk, and it is widely used in baking and in commercial milk used for infants.

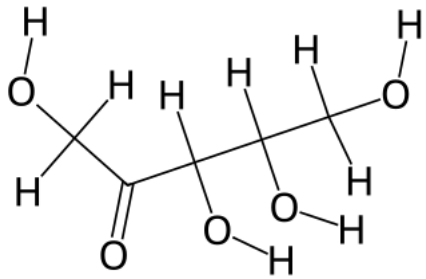
Fructose has the molecular formula $C_6H_{12}O_6$, the same as glucose, so the two are isomers. Sucrose, another important carbohydrate, is present in sugar cane and sugar beets, and sucrose, lactose and maltose all have the molecular formula $C_{12}H_{22}O_{11}$.

25.2 Draw open chain structures of glucose and fructose.

Glucose (aldohexose): its open chain structure contains an aldehyde group (–CHO) at carbon 1 and five hydroxyl groups (–OH), so its condensed open chain formula is $CHO–(CHOH)_4–CH_2OH$. Its molecular formula is $C_6H_{12}O_6$.



Fructose (ketohexose): its open chain structure contains a ketonic group (>C=O) at carbon 2 and five hydroxyl groups, so its condensed open chain formula is $CH_2OH–CO–(CHOH)_3–CH_2OH$. Its molecular formula is also $C_6H_{12}O_6$.



Both compounds have the same molecular formula but different functional groups, so they are structural isomers. Glucose is an aldose and fructose is a ketose.

25.3 Construct peptide linkage and explain the structure of a polypeptide.

An amino acid has two functional groups, a carboxyl group (–COOH) and an amino group ($–NH_2$). The basic reaction which leads to the formation of long chains of proteins involves the elimination of water from two molecules of the same or different amino acids. The carboxyl group of one amino acid reacts with the amino group of the other, water is removed, and a linkage –CO–NH– is formed. This linkage is called the peptide linkage.

$$H_2N–CHR–COOH + H_2N–CHR'–COOH \rightarrow H_2N–CHR–CO–NH–CHR'–COOH + H_2O$$

The molecule formed by the joining of two amino acids is called a peptide. It still has a carboxyl group and an amino group at its two ends, so it can react further with more amino acids to give a long chain called a polypeptide. This long chain then undergoes twists and turns and assumes a three dimensional structure, which is the structure of a protein. All the thousands of proteins of the human body are formed by linking only twenty amino acids in this way; the simplest amino acid is glycine.

25.4 What are DNA and RNA? Describe the role played by them in our body.

DNA is deoxyribonucleic acid and RNA is ribonucleic acid. They are nucleic acids, which are responsible for the existence, development and reproduction of all forms of life. Both are chemically similar compounds but they play different roles within the cell.

DNA: molecules of DNA are very large and are mostly found in the nucleus of the cells. The function of DNA is to store information and pass it on to RNA at the proper time, so DNA carries the hereditary characters from parents to children.

RNA: RNA molecules are much smaller and are mostly found outside the cell nucleus. The function of RNA is to read, decode and use the information of DNA to make proteins, which are needed for growth and repair of the body.

25.5 Compare the roles played by carbohydrates, proteins and lipids in our body.

Carbohydrates	Proteins	Lipids
They are the main source of energy for the body.	They build new cells and repair the injured ones.	They act as major energy reserves of the body.
They help the body to use protein and fat efficiently.	They act as enzymes, hormones and haemoglobin.	They provide insulation and protection for vital organs.
A healthy adult needs about 225–325 g per day.	An adult should consume about 60 g per day.	A daily intake of fats is about 44–78 g.
Examples: glucose, starch and cellulose.	Examples: enzymes, insulin and haemoglobin.	Examples: fats, oils and triglycerides.

Lipids provide, on the average, twice as much energy as provided by carbohydrates or proteins, but carbohydrates are the cheapest and the quickest source of energy.

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